An Exact Stability, Crossing, and Jordan Atlas for Erlang–2 Distributed-Delay Feedback

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Abstract

We solve a scalar negative-feedback equation with a normalized Erlang shape-two memory kernel. Compatible histories admit an exact three-dimensional linear-chain realization, whose cubic Routh determinant gives a necessary-and-sufficient stability threshold and a transverse imaginary crossing. We further exhaust every repeated-root surface, determine all Jordan sizes, count unstable roots, and give closed semigroup formulas on simple, double, and triple-root faces.

1 Frozen memory owner

Fix

$$a, b \geq 0, \qquad r > 0, \qquad K_r(s) = r^2 s \mathrm{e}^{-rs} \quad (s \geq 0),$$

and consider

$$\dot{x}(t) = -ax(t) - b\int_0^\infty K_r(s)x(t-s)\,ds. \tag{1}$$

The kernel is normalized; it has mean $2/r$, variance $2/r^2$, and Laplace transform $r^2/(\lambda+r)^2$. A compatible fading-memory history is one for which the following integrals exist and permit the standard differentiation under the integral:

$$z_1(t) = \int_0^\infty r\mathrm{e}^{-rs}x(t-s)\,ds, \qquad z_2(t) = \int_0^\infty K_r(s)x(t-s)\,ds.$$

Lemma 1 (Exact linear chain)**.** *For compatible initialization, equation (1) is equivalent to*

$$\dot{x} = -ax - bz_2, \qquad \dot{z}_1 = r(x - z_1), \qquad \dot{z}_2 = r(z_1 - z_2). \tag{2}$$

It is not asserted that arbitrary values of (z_1, z_2) encode a prescribed prehistory.

Proof. Integration by parts gives the two filter equations in (2); the first equation is (1). Conversely, subtract the displayed convolutions from a solution of the two stable filter equations. The differences solve homogeneous first-order equations and vanish initially, so they vanish for all later times. Thus compatible chain solutions recover the delay equation exactly. □

Put $X = (x, z_1, z_2)^\top$. Then $\dot{X} = \mathsf{M}X$, where

$$\mathsf{M} = \begin{pmatrix} -a & 0 & -b \\ r & -r & 0 \\ 0 & r & -r \end{pmatrix},$$

and direct expansion gives

$$\begin{aligned} p(\lambda) = \det(\lambda I - \mathsf{M}) &= (\lambda + a)(\lambda + r)^2 + br^2 \\ &= \lambda^3 + c_2\lambda^2 + c_1\lambda + c_0, \\ c_2 = a + 2r, \qquad c_1 &= r(r + 2a), \qquad c_0 = r^2(a + b). \end{aligned} \tag{3}$$

Theorem 2 (Sharp stability and imaginary crossing)**.** *For $b > 0$, the zero solution is exponentially stable if and only if*

$$0 < b < b_H, \qquad b_H = \frac{2(a+r)^2}{r}. \tag{4}$$

At $b = b_H$,

$$p(\lambda) = (\lambda + a + 2r)(\lambda^2 + r(r + 2a)), \tag{5}$$

so the roots are $-a - 2r$ and $\lambda_\pm = \pm\mathrm{i}\omega_H$, with $\omega_H = \sqrt{r(r+2a)}$. The conjugate pair is simple and crosses from left to right:

$$\operatorname{Re}\lambda_+'(b_H) = \frac{r^2}{2\{r(r+2a) + (a+2r)^2\}} > 0. \tag{6}$$

For $b > b_H$, exactly two roots lie in the open right half-plane.

Proof. For $b > 0$, all c_j are positive. The remaining cubic Routh–Hurwitz minor is

$$c_2 c_1 - c_0 = r\{2(a+r)^2 - br\}.$$

This proves the equivalence in (4). At equality, multiplication verifies (5). Implicit differentiation of $p(\lambda, b) = 0$ at $\lambda = \mathrm{i}\omega_H$ yields (6). Above the threshold the first Routh column has signs $+, +, -, +$, hence two sign changes and exactly two open-right-half-plane roots. □

The phrase *chain equivalence owner* records the first manuscript round: this is an all-parameter analytic theorem, not stability inferred from sampled roots.

2 Complete degeneracy and propagator atlas

Theorem 3 (All repeated roots and Jordan blocks)**.** *In the domain $a, b \geq 0$, $r > 0$, all repeated-root cases are precisely:*

1. *$b = 0$. The root $-r$ has one size-two Jordan block when $a \neq r$, and at $a = r$ the root $-r$ has one size-three block.*
2. *$0 \leq a < r$ and*

$$b = b_D = \frac{4(r-a)^3}{27r^2} > 0. \tag{7}$$

Here

$$p(\lambda) = (\lambda - \mu)^2(\lambda - \nu), \qquad \mu = -\frac{r+2a}{3}, \quad \nu = -\frac{4r-a}{3},$$

and the Jordan sizes are two and one.

The positive-b defective face lies strictly inside the stable region.

Proof. Differentiating (3) and taking the discriminant gives

$$p'(\lambda) = (\lambda + r)(3\lambda + r + 2a), \qquad \operatorname{disc} p = br^2\{4(r-a)^3 - 27br^2\}. \tag{8}$$

The first critical root belongs to p exactly when $b = 0$. The second belongs to p exactly on (7); it has positive b precisely when $a < r$. These observations exhaust (8), and substitution gives the factorization. Moreover,

$$\det[e_1, \mathsf{M}e_1, \mathsf{M}^2 e_1] = r^3 \neq 0.$$

Thus M is cyclic and its minimal polynomial equals p: each repeated root has one Jordan block of its algebraic multiplicity. Finally

$$\frac{b_D}{b_H} = \frac{2(r-a)^3}{27r(a+r)^2} < 1,$$

so the positive defective face is stable. □

For distinct roots $\lambda_1, \lambda_2, \lambda_3$, Lagrange interpolation modulo p gives

$$\mathrm{e}^{t\mathsf{M}} = \sum_{j=1}^{3} \mathrm{e}^{t\lambda_j} \prod_{k \neq j} \frac{\mathsf{M} - \lambda_k I}{\lambda_j - \lambda_k}. \tag{9}$$

On a double face, write $p = (\lambda - \mu)^2(\lambda - \nu)$ and set

$$P_\nu = \frac{(\mathsf{M} - \mu I)^2}{(\nu - \mu)^2}, \qquad P_\mu = I - P_\nu, \qquad N_\mu = (\mathsf{M} - \mu I)P_\mu.$$

Hermite interpolation yields

$$\mathrm{e}^{t\mathsf{M}} = \mathrm{e}^{\mu t}(P_\mu + tN_\mu) + \mathrm{e}^{\nu t}P_\nu, \qquad N_\mu^2 = 0. \tag{10}$$

At $a = r, b = 0$, $N = \mathsf{M} + rI$ satisfies $N^3 = 0$ and

$$\mathrm{e}^{t\mathsf{M}} = \mathrm{e}^{-rt}(I + tN + t^2N^2/2). \tag{11}$$

Equations (9)–(11) close the flow on every spectral stratum.

The phrase *Routh Hopf atlas owner* marks the substantive first revision: transversality, unstable-root count, discriminant exhaustion, Jordan sizes, and defective semigroups have all been added.